

UNIVERSITY OF ENGINEERING & TECHNOLOGY, TAXILA



Term Project

Filament X: An Approach towards Sustainability

Submitted To

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Course

ENTREPRENEURSHIP

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Executive Summary

The project report outlines the establishment of **Filament X**, a Pakistani company aimed at revolutionizing the 3D printing industry by providing affordable, high-quality PLA (Polylactic Acid) filaments. Addressing critical challenges such as high costs (ranging from PKR 4,000 to 8,500 per kilogram) and inadequate recycling systems for PLA waste, Filament X introduces a circular economy model that recycles waste PLA into new filaments priced at just PKR 3356.01 PKR/kg. Targeting students and small to large businesses, the company emphasizes sustainability and innovation in its operations. Filament X's marketing strategy includes partnerships with educational institutions and local retailers to enhance accessibility and promote eco-friendly practices. With a strong commitment to growth and environmental responsibility, Filament X aims to lead the market in sustainable 3D printing solutions.

What sets Filament X apart is our commitment to:

- **Sustainability:** By recycling waste PLA, we're creating a cleaner future while reducing material waste.
- **Affordability:** Offering premium filaments at prices that make 3D printing more accessible.
- **Innovation:** Using smart reverse-engineering techniques to cut costs and optimize resources.

To spread the word, we're working closely with universities, partnering with influencers to highlight our eco-friendly mission, and running workshops to teach people about sustainability in 3D printing.

With a lean, efficient team structure, we're built for fast decision-making and steady growth. Financially, we're in a great position: we'll hit our break-even point after selling just 266 units/month, which means we can start generating profits quickly.

Filament X isn't just about making filaments; it's about leading a change. By combining sustainability with innovation, we're building a business model that's ready to scale, connect with new partners, and help shape the future of 3D printing in Pakistan

Acknowledgment

We would like to express our heartfelt gratitude to everyone who contributed to the success of this project report on Filament X. Special thanks to our Course Instructor Mr. Umair Shahid for his invaluable guidance and support throughout the development process. We are also grateful to our team members for their dedication and hard work, which were crucial in bringing this vision to life. Our appreciation extends to the educational institutions and local businesses that partnered with us, helping us understand market needs and enhance our outreach.

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1. Problem Statement

The 3D printing industry in Pakistan faces significant challenges, including limited access to affordable, high-quality PLA filaments and the lack of an effective recycling system for PLA waste. Existing suppliers offer filaments at high costs ranging from Rs. 4,000 to Rs. 8,500 per kilogram, making them less accessible to students, small businesses, and industries. Additionally, the improper disposal of PLA waste exacerbates environmental concerns, as most waste materials are discarded rather than repurposed. This creates a critical need for a sustainable and cost-effective solution that addresses both the demand for affordable filaments and the recycling of PLA waste to promote environmental sustainability.

2. Company Description

Filament X is a pioneering Pakistani company that is redefining the 3D printing industry through innovation, sustainability, and customer-focused solutions. Specializing in the production of high-quality, eco-friendly PLA (Polylactic Acid) filaments, and the company combines affordability with environmental responsibility to meet the growing demands of the 3D printing market. Filament X is not just about creating products—it's about creating a sustainable future. By embedding a circular economy model into its operations, the company actively reduces waste, optimizes resources, and sets a new standard for responsible manufacturing.

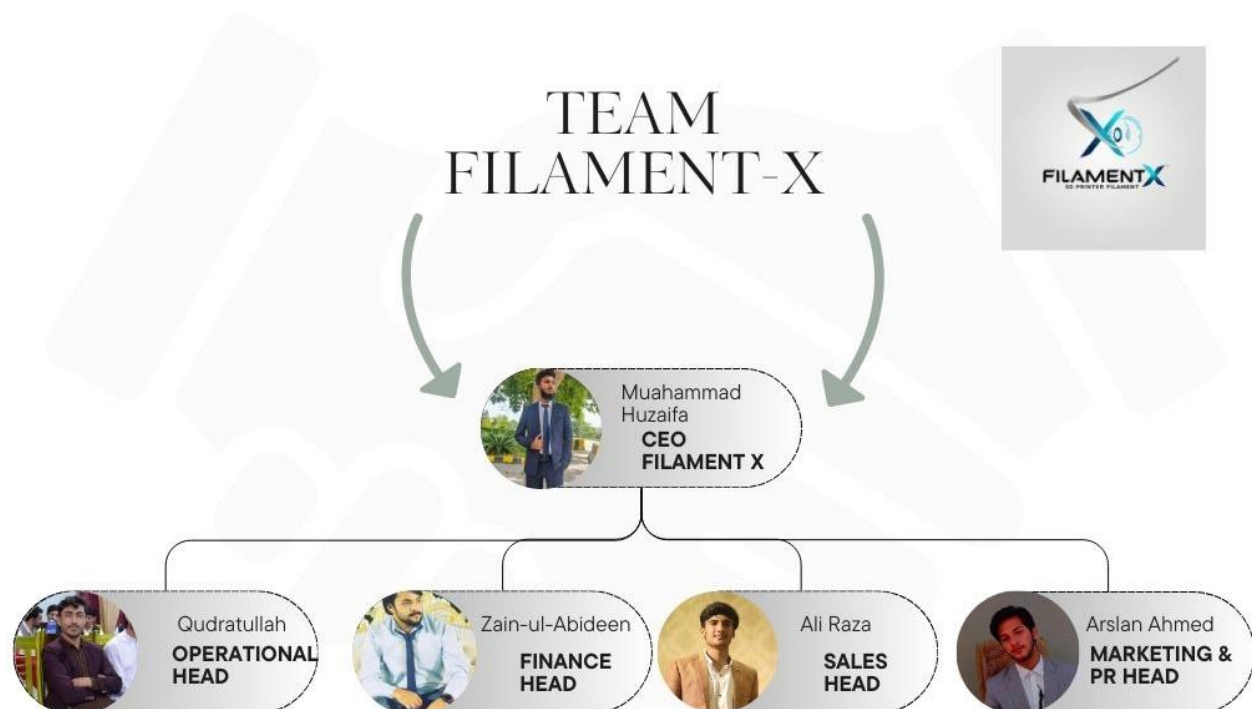


Figure 1: Team Filament X

2.1 Mission and Vision

At the heart of Filament X is a clear mission: to revolutionize 3D printing by providing biodegradable, innovative materials that inspire creativity and empower industries. The company's aim is to make sustainable 3D printing accessible to everyone, from students and educators to businesses and manufacturers. Filament X envisions a future where it leads the global market in sustainable 3D printing materials, becoming a benchmark for environmental preservation, technological advancement, and superior quality. This vision drives every aspect of the business, pushing boundaries to deliver products that seamlessly integrate sustainability and innovation.

2.2 Sustainability and Operations

Filament X operates on the principles of sustainability and efficiency. Its unique recycling initiative allows the company to repurpose used PLA materials into new, premium-quality filaments, reducing waste and carbon emissions while conserving natural resources. By sourcing high-grade PLA pellets from eco-conscious suppliers and using locally procured materials, Filament X not only ensures the highest product standards but also supports local businesses and reduces supply chain costs. The streamlined inventory system minimizes waste and guarantees a consistent supply to meet diverse customer needs, making Filament X a trusted name in the 3D printing industry.

The company's operations are designed to cater to a wide range of customers. Students and researchers benefit from affordable filaments for academic projects, while small businesses find reliable and cost-effective solutions for their entrepreneurial ventures. For large enterprises, Filament X provides industrial-grade filaments optimized for bulk production, ensuring compliance with environmental and quality standards. This ability to serve varied needs, combined with competitive pricing, has positioned Filament X as a leader in the market.

2.3 Commitment to Growth and Innovation

Filament X is committed to continuous growth and innovation. The company plans to expand its product offerings to cater to emerging industry demands and invest in cutting-edge research and development to stay ahead of technological advancements. With a focus on entering international markets, Filament X aims to bring its sustainable and innovative 3D printing solutions to a global audience. By staying true to its mission and values, Filament X is not just shaping the future of 3D printing but also contributing to a greener, more sustainable world.

Through its dedication to sustainability, innovation, and customer satisfaction, Filament X is setting a new benchmark in the 3D printing industry. It is not just a company—it is a movement toward a more responsible and creative future.

3. Market Positioning

Filament X has carved a unique position for itself as a forward-thinking and sustainable provider in the global 3D printing industry. By addressing critical gaps in affordability, environmental impact, and product quality, the company has established a strong competitive edge. Specializing in recycled Polylactic Acid (PLA) filaments, Filament X is not just meeting the needs of its diverse customer base but also actively contributing to global sustainability goals. This dual focus on innovation and responsibility allows the company to stand out in a rapidly evolving market.

3.1 Target Market and Customer Focus

Filament X caters to three primary customer groups, each with distinct needs. Educational institutions, including universities, colleges, and schools, form a significant portion of the customer base. These organizations, often engaged in STEM education and maker-space activities, benefit from Filament X's affordable, eco-friendly filaments for research, prototyping, and teaching. By offering sustainable materials, Filament X aligns perfectly with the environmental values embraced by many educational institutions.

Small and medium enterprises (SMEs) are another key focus for Filament X. Industries like manufacturing, prototyping, and design require reliable and cost-effective materials to maintain competitive production standards. Filament X's recycled filaments provide these businesses with an economical alternative to traditional materials, enabling them to reduce costs without sacrificing quality. For large-scale enterprises in industries such as aerospace, automotive, and healthcare, Filament X offers tailored solutions designed to meet rigorous demands. These businesses, which use 3D printing for advanced applications like tooling, custom components, and medical devices, can achieve their corporate social responsibility goals with the help of Filament X's eco-friendly products.

3.2 Competitive Edge and Market Presence

Filament X's competitive advantage lies in its ability to integrate sustainability, affordability, and

quality seamlessly. The company's use of recycled PLA in its production process significantly reduces waste and environmental impact, meeting the growing demand for sustainable 3D printing materials. At the same time, Filament X offers its products at prices significantly lower than those of competitors, ensuring accessibility to a wide audience. Despite the competitive pricing, the company guarantees high-quality filaments, leveraging advanced reverse engineering techniques to ensure their performance is on par with materials made from virgin PLA.

To enhance its market presence, Filament X employs a robust and diversified distribution strategy. The company's e-commerce platform is user-friendly, SEO-optimized, and designed to simplify the purchasing process for customers around the world. By collaborating with educational institutions and research facilities, Filament X builds long-term relationships and ensures bulk sales, reinforcing its presence in the academic and research sectors. Additionally, partnerships with local and regional retailers strengthen the company's physical availability and brand visibility, creating a well-rounded market outreach.

Through its thoughtful approach to market positioning and unwavering commitment to sustainability and innovation, Filament X is successfully bridging the gap between affordability, environmental responsibility, and product excellence. The company's strategic focus on diverse customer needs and its proactive market engagement ensures a strong and lasting impact on the 3D printing industry.

4. Marketing Strategy

Filament X's marketing strategy is rooted in sustainability, innovation, and meaningful customer engagement. The company aims to establish itself as a leader in eco-friendly 3D printing solutions by adopting targeted outreach, strong branding, and a well-rounded approach to connecting with its diverse audience. By aligning its marketing efforts with its core mission and values, Filament X seeks to not only drive sales but also create a lasting impact on the 3D printing industry while building loyalty among customers who share its vision of a sustainable future.

4.1 Building the Filament X Brand

Filament X positions itself as a premium, eco-conscious brand that delivers high-quality recycled PLA filaments at affordable prices. This combination of sustainability and value sets it apart from competitors and appeals strongly to environmentally conscious individuals and organizations. The emphasis on protecting the planet while offering innovative solutions resonates deeply with

customers who prioritize eco-friendly choices in their purchasing decisions. By consistently promoting this unique positioning, Filament X builds trust and credibility as a forward-thinking, responsible brand.

4.2 Reaching the Audience through Strategic Channels

Filament X employs a variety of marketing channels to connect with its audience and amplify its message. A robust digital marketing strategy forms the foundation of its efforts, leveraging platforms like Instagram, Facebook, and LinkedIn to engage customers with educational content, product showcases, and customer success stories. The company also uses content marketing—such as blogs, tutorials, and videos—to highlight the benefits of recycled PLA and demonstrate how its products can drive innovation in 3D printing. Its SEO-optimized, user-friendly e-commerce website further ensures a seamless shopping experience, driving online sales and enhancing customer satisfaction.

Partnerships play a crucial role in expanding Filament X's reach and credibility. Collaborations with universities and educational institutions allow the company to host workshops, provide discounted bulk supplies, and build long-term relationships with students and researchers. Engaging with influencers and thought leaders helps to promote Filament X's sustainability message to a broader audience, while partnerships with local retailers increase physical availability and strengthen brand awareness. Participation in events like ITCN Asia and the Pakistan Tech Expo positions Filament X as a leader in the 3D printing space, showcasing its innovations and commitment to sustainability on a prominent stage.

Filament X also prioritizes community engagement as part of its marketing strategy. By hosting free 3D printing workshops at universities and maker spaces, the company educates the public on sustainable practices and the benefits of recycling. Sponsoring research projects and student initiatives further embeds Filament X in the academic ecosystem, fostering goodwill and loyalty among young innovators who represent the future of the industry.

4 MAIN MARKETING STRATEGY FOLLOWED BY FILAMENT-X



Figure 2: 4 Marketing Strategies

4.3 Engaging Customers with Targeted Campaigns

Filament X runs targeted campaigns to highlight its key selling points. For sustainability-focused customers, the company emphasizes the environmental benefits of using recycled PLA filaments, showcasing how these products contribute to reducing waste and promoting a circular economy. Cost-efficiency campaigns appeal to SMEs and hobbyists, demonstrating how Filament X delivers affordable, high-quality materials that rival those of competitors. For advanced industries like healthcare, automotive, and aerospace, the company shares case studies and testimonials to illustrate the innovative applications of its products in real-world scenarios.

To retain customers and build lasting relationships, Filament X invests in loyalty programs, offering discounts, early access to new products, and exclusive deals for repeat buyers. Personalized marketing ensures that email campaigns and promotions are tailored to individual purchasing histories and preferences, creating a more engaging and relevant experience. Exceptional customer service is another cornerstone of the strategy, with Filament X prioritizing prompt, helpful responses to customer inquiries to foster trust and satisfaction.

4.4 Sustainability as the Heart of Marketing

Sustainability is at the core of Filament X's marketing strategy, woven into every aspect of its outreach. By consistently emphasizing the environmental benefits of recycled PLA and highlighting the company's circular economy model, Filament X reinforces its commitment to protecting the planet. This focus on sustainability not only appeals to eco-conscious customers but also positions the company as a socially responsible leader in the industry. By staying true to its mission and values, Filament X is creating a brand identity that resonates deeply with its audience and drives lasting success in the 3D printing market.

5. Sales Plan

The Filament X sales plan is a strategic blueprint for achieving market leadership in the 3D printing industry by delivering value through sustainable, high-quality, and affordable products. This plan is rooted in understanding customer needs, leveraging diverse sales channels, and fostering long-term relationships. By prioritizing accessibility, scalability, and customer engagement, Filament X positions itself as the go-to provider of eco-friendly 3D printing materials.

5.1 Strategic Sales Plan

Filament X aims to lead the 3D printing industry by delivering affordable, sustainable, and high-quality PLA filaments. The company's sales objectives focus on achieving a monthly target of 500 kilograms within the first year, generating steady revenue, and building enduring customer relationships through superior service and loyalty programs. By promoting the environmental benefits of recycled PLA filaments, Filament X strengthens its reputation as a leader in eco-friendly solutions while addressing the needs of educational institutions, SMEs, and large enterprises.

5.2 Multi-Channel Approach and Target Markets

Filament X employs a diverse sales strategy to reach its target audience. A modern e-commerce platform ensures seamless global access, while collaborations with universities and research facilities foster long-term partnerships and bulk sales. The company's presence in retail and maker spaces enhances availability for small businesses and hobbyists. Participation in industry events such as ITCN Asia further expands visibility and creates opportunities for high-value contracts. Target markets include educational institutions seeking affordable materials, SMEs prioritizing

cost-effective solutions, and large enterprises needing high-quality, scalable supplies.

5.3 Customer Retention and Growth Metrics

To retain customers and ensure satisfaction, Filament X offers exceptional after-sales support, loyalty programs, and personalized services. The company focuses on meeting customer needs with a competitive pricing strategy of PKR 3356.01 PKR/kg, bulk discounts, and promotional campaigns. Sales performance is tracked using metrics such as monthly revenue, sales volume, and customer retention rates, with a long-term goal of entering international markets within three years. By aligning its sales plan with its sustainability mission, Filament X ensures continued growth while setting new standards in the 3D printing industry.

6. Competitive Analysis

The 3D printing filament market is dynamic and rapidly growing, with numerous competitors offering a variety of products to meet increasing consumer demand. Filament X stands out with its innovative focus on sustainability, affordability, and quality, addressing key industry challenges and catering to an eco-conscious customer base. This analysis explores the competitive landscape, highlighting Filament X's strengths, opportunities, and the potential challenges it faces.

Filament X operates in a market valued at \$1.06 billion in 2023, projected to grow to \$2.3 billion by 2031 at a compound annual growth rate (CAGR) of 12.8%. This expansion is driven by the increasing demand for eco-friendly and high-performance materials, presenting both opportunities and hurdles for a company like Filament X entering a competitive environment.

6.1 Filament X's Unique Market Position

Filament X has carved out a strong competitive edge by emphasizing sustainability, affordability, and customer satisfaction. Its recycled PLA filaments align with the global push for environmentally friendly products, positioning it as a leader in sustainable practices. This commitment resonates with eco-conscious consumers and organizations, distinguishing Filament X in a market increasingly focused on responsible production.

Affordability is another key strength. By pricing its products at PKR 3356.01 PKR/kg, significantly below market averages, Filament X makes high-quality filaments accessible to a wide range of customers, including educational institutions, small businesses, and hobbyists. Its

innovative recycling practices, which include a circular economy model that repurposes waste PLA, further set the company apart by reducing environmental impact and optimizing resources. Filament X's customer-centric approach, which involves tailored solutions, workshops, and exceptional support, fosters strong relationships and long-term loyalty, solidifying its position in the market.

6.2 Opportunities for Growth

The growing demand for sustainable materials creates ample opportunities for Filament X to expand into international markets. By strengthening collaborations with universities and schools, the company can secure bulk sales and enhance brand recognition. Additionally, diversifying its product offerings to include other types of filaments or 3D printing accessories could help Filament X attract an even broader customer base, ensuring sustained growth and relevance in the industry.

6.3 Potential Threats

Despite its strengths, Filament X faces challenges in an increasingly competitive landscape. Established players with larger budgets and global reach could introduce similar eco-friendly products, intensifying market competition. Supply chain disruptions pose another risk, as consistent availability of raw materials is crucial for maintaining production efficiency and quality. Furthermore, the price-sensitive nature of the industry may lead competitors to undercut prices, requiring Filament X to remain agile and strategic in maintaining its competitive edge.

By leveraging its strengths, addressing challenges proactively, and capitalizing on growth opportunities, Filament X is well-positioned to thrive in the expanding 3D printing filament market. Its innovative approach to sustainability and affordability ensures it remains a standout choice for customers seeking reliable and eco-conscious solutions.

7. Organizational Structure

7.1 Core Team:

Each team member holds a **bachelor's degree in industrial engineering** and brings fresh, innovative perspectives to the business. Though recent graduates, their academic projects and internships have equipped them with relevant skills to drive the company forward.

1. CEO – Muhammad Huzaifa:

- Role: Strategic leadership and decision-making.
- Qualification: B.S. Industrial Engineering, specialized in Operations Strategy and Project

Management.

- Experience: Led multiple academic projects involving process optimization and lean manufacturing during studies.
- 2. Operations Head - Qudrat Ullah:**
 - Role: Overseeing production, supply chain, and logistics.
 - Qualification: B.S. Industrial Engineering, specialized in Manufacturing Processes.
 - Experience: Internship in production management at a local manufacturing firm.
- 3. Marketing & PR Head - Arslan Ahmad:**
 - Role: Branding, promotions, and public relations.
 - Qualification: B.S. Industrial Engineering, with additional coursework in Marketing Strategies.
 - Experience: Academic exposure to marketing analytics and a research project on customer engagement.
- 4. Sales Head – Ali Raza:**
 - Role: Customer acquisition, retention, and sales growth.
 - Qualification: B.S. Industrial Engineering, specialized in Sales Optimization.
 - Experience: Practical experience in customer relationship management through internship work.
- 5. Finance Head - Zain Ul Abideen:**
 - Role: Financial planning, budgeting, and cost management.
 - Qualification: B.S. Industrial Engineering, with additional training in Financial Modeling.
 - Experience: Internship in financial analysis at a medium-scale enterprise.

7.2 Support Staff:

If a company feel the need to hire any employ then the team will decide and will hire employ skilled operators for machinery, logistics personnel for supply chain management, and inspectors for quality control. Training will ensure all staff align with Eco Taxila Filaments' operational standards. (In the current situation these five members will perform all operations)

8. Products and Services

Eco Taxila Filaments offers a diverse range of high-quality, sustainable PLA filament solutions tailored to meet the needs of its target customers in education, industry, and small-scale production.

8.1 Product Offerings

- 1. Standard PLA Filaments:**
 - Biodegradable, eco-friendly filaments designed for general-purpose 3D printing.
 - Available in customizable colors and properties to cater to diverse customer needs.
- 2. Specialty Filaments:**
 - Future plans include high-strength, heat-resistant, and specialty filaments to support

advanced and industrial-grade applications.

3. Recycling Services:

- Buy-back programs for PLA waste from customers.
- Reprocessing waste into reusable, high-quality filaments, promoting a circular economy and reducing environmental impact.

8.2 Unique Selling Points

1. Locally Manufactured:

- Eliminates reliance on imports, reducing costs and ensuring faster delivery.

2. Eco-Friendly Solutions:

- Emphasizes sustainability through biodegradable products and recycling initiatives, appealing to environmentally conscious buyers.

3. Customization and Versatility:

- Offers flexibility in filament properties and colors to meet specific customer requirements.

4. Competitive Pricing:

- Filaments priced at PKR 5,356/kg, undercutting imported alternatives without compromising quality.

5. Faster Delivery:

- Local production ensures reduced lead times, meeting urgent customer demands

9. Operating Plan

Filament X operates from a well-organized, cost-efficient facility designed to support both production and administrative tasks. The 1,000 sq. ft. space integrates a dedicated production area and an office setup to ensure smooth operations. The production area houses extrusion machines, cooling systems, and spooling equipment, while the office area is used for administrative tasks, inventory management, and staff coordination. With a monthly rent of PKR 10,000, the facility is optimized for both functionality and affordability.

9.1 Streamlined Production Workflow

Filament X's production process is carefully designed to deliver high-quality PLA filaments consistently and efficiently. The workflow begins with sourcing premium raw PLA resin, which is procured locally from Karachi and internationally through platforms like Alibaba. To ensure uninterrupted production, the company maintains a 30-day buffer stock in climate-controlled storage to preserve resin quality.

The extrusion process is powered by two machines, each with a capacity of 1 kg/hour, allowing for a monthly output of up to 320 kg. The melted resin is shaped into filaments and cooled using a dedicated system, ensuring consistent diameter and strength. The cooling process is supported by a reliable water supply infrastructure. Once solidified, the filaments are wound onto durable spools using automated spooling machines, making them ready for transport and storage. Quality inspection is a critical step, where filaments are tested for diameter consistency, smoothness, and strength using advanced measurement tools. Only those meeting the highest standards are packaged in secure materials, including labeled boxes and shrink wrap, for professional and safe delivery.

9.2 Inventory Management

To maintain a steady production flow, Filament X employs efficient inventory management practices. A 30-day buffer stock of raw materials, including PLA resin, additives, and spools, is stored in climate-controlled conditions to ensure quality. Finished products are managed using a FIFO (First-In, First-Out) system, which minimizes aging inventory and maintains freshness for customers. This approach ensures that operations are not disrupted and that customers receive high-quality products every time.

9.3 Utilities and Safety

Ensuring uninterrupted operations and safety is a top priority at Filament X. The facility is equipped with a stable electricity supply and a backup system, set up at a cost of PKR **1,165,160**, to prevent any downtime. A water supply system, installed at a cost of PKR 56,000, supports the cooling operations crucial to filament production. To create a comfortable and safe working environment, PKR 28,000 has been invested in proper lighting and ventilation. Fire safety measures, including extinguishers and alarms, are in place to comply with safety standards and protect both staff and equipment.

By maintaining a focus on efficiency, quality, and safety, Filament X ensures that its operations are not only reliable but also aligned with its commitment to sustainability and customer satisfaction. The well-thought-out operating plan supports the company's mission to provide eco-friendly, high-quality 3D printing materials.

10. Financial Plan

The financial plan outlines the cost structure, revenue projections, and profitability for Eco Taxila Filaments, reflecting the data provided.

10.1 Fixed Costs

Table 1: Fixed Cost

These are one-time setup costs that remain constant regardless of production quantity.			
Item	Quantity	Unit Cost (PKR)	Total Cost (PKR)
Extrusion Machines	2	305,580	611,160
Cooling System	1	56,000	56,000
Spooling Machines	2	108,000	216,000
Quality Control Tools	1 package	14,000	14,000
Electricity Setup and Backup	1 setup	280,00	280,00
Water Supply Installation	1 setup	56,000	56,000
Lighting and Ventilation	1 system	8,000	8,000
Office Furniture	1 setup	14,000	14,000
Storage Racks	1 batch	30,000	30,000
Fire Safety Equipment	1 package	28,000	28,000
IT Setup (Computer, Printer, etc.)	1 setup	132,000	132,000
Total Fixed Costs			1,165,160

10.2 Variable Costs

These costs vary directly with production quantity.

Table 2: Variable Cost

Item	Cost per kg (PKR)	Monthly Cost (PKR)	Remarks
Raw PLA Resin	1,003.20	351,120	Based on 320 kg production
Additives (Colorants, etc.)	215.00	75,250	
Spools	30.00	14,400	1 spool/kg @ PKR 30.00
Packaging Materials	448.00	156,800	
Salaries	-	175,000	5 staff @ PKR 50,000/month
Utilities (Electricity, Water, etc.)	-	60,000	Estimated operational costs
Maintenance Costs	-	20,000	Routine maintenance
Miscellaneous Expenses	-	10,000	Contingency fund
Total Variable Costs	-	862570.00	

Monthly Production Analysis for 320 kg

10.3 Total Monthly Costs

The total monthly cost includes fixed costs (spread over 36 months) and updated variable costs.

1. Fixed Costs per Month

Fixed costs are spread across 3 years (36 months):

$$\text{Monthly Fixed Costs} = \text{Total Fixed Costs} / 36 = 1,165,160 / 36 = 32,365 \text{ PKR}$$

Variable Costs Per Month Salaries have been updated to 175,000 PKR (5 workers at 35,000 each). Total Variable Costs = 862570.00 PKR

$$\text{Total Monthly Costs} = \text{Monthly Fixed Costs} + \text{Variable Costs}$$

$$\text{Total Monthly Costs} = 32,365 + 862570.00 = 894935.56 \text{ PKR}$$

2. Cost Price per kg

The cost price per kg is calculated by dividing the total monthly cost by the monthly production quantity.

$$\text{Cost Price per kg} = \text{Total Monthly Costs} / \text{Monthly Production}$$

$$\text{Cost Price per kg} = 894935.56 / 320 = 2796.67 \text{ PKR/kg}$$

this means it costs 2796.67 PKR to produce 1 kg of filament when producing 320 kg in a month.

3. Selling Price per kg

The selling price is calculated by adding a 20% profit margin to the cost price.

$$\text{Selling Price per kg} = \text{Cost Price per kg} \times (1 + 0.2)$$

$$\text{Selling Price per kg} = 2796.67 \times 1.2 = 3356.01 \text{ PKR/kg}$$

This means each kg will be sold for 3356.01 PKR.

4. Break-Even Point

The break-even point is the production level at which revenue equals costs. It shows how many kg need to be sold to cover all costs.

$$\text{Break-Even Production (kg)} = \text{Total Monthly Costs} / \text{Selling Price per kg}$$

$$\text{Break-Even Production (kg)} = 894935.56 / 3356.01 \approx 266.67 \text{ kg}$$

This means you need to produce and sell approximately 266.67 kg per month to cover all costs.

5. Monthly Profit

If the production is 320 kg, calculate the revenue and subtract the total monthly costs to find profit.

1. Revenue

$$\text{Revenue} = \text{Selling Price per kg} \times \text{Monthly Production}$$

$$\text{Revenue} = 3356.01 \times 320 = 1073922.67 \text{ PKR}$$

2. Profit

$$\text{Profit} = \text{Revenue} - \text{Total Monthly Costs}$$

$$\text{Profit} = 1073922.67 - 894935.56 = 178987.11 \text{ PKR}$$

this means the business earns 178987.11 PKR per month at 320 kg production.

Summary

- Cost Price per kg: 2796.67 PKR
- Selling Price per kg: 3356.01 PKR Break-Even Production: 266.67 kg
- Monthly Profit (at 320 kg production): 178987.11 PKR

11. Funding Requirements

Eco Taxila Filaments will be funded entirely through **bootstrapping**, with the founders leveraging personal savings and private contributions to cover initial startup costs and operational expenses. This approach ensures full ownership, operational control, and a lean financial model.

11.1 Bootstrapping Strategy

1. Initial Capital:

- The total startup cost of PKR 2027730 will be funded through pooled personal savings and contributions from the founding team.
- A portion of the funds will be allocated to cover early operational expenses, including raw material procurement and facility setup.

2. Operational Sustainability:

- Profit generated from monthly production and sales (178987.11 PKR per month at 320 kg production) will be reinvested to cover variable costs, recurring expenses, and incremental growth.
- No external debt or equity funding will be utilized, reducing financial liabilities.

3. Growth and Expansion:

- Future scaling, such as increasing production capacity or diversifying product offerings, will be financed through retained earnings and operational profits.

12. Appendix

The appendix includes detailed supplementary information to support the business plan for Eco Taxila Filaments.

12.1 Detail Cost Break Down

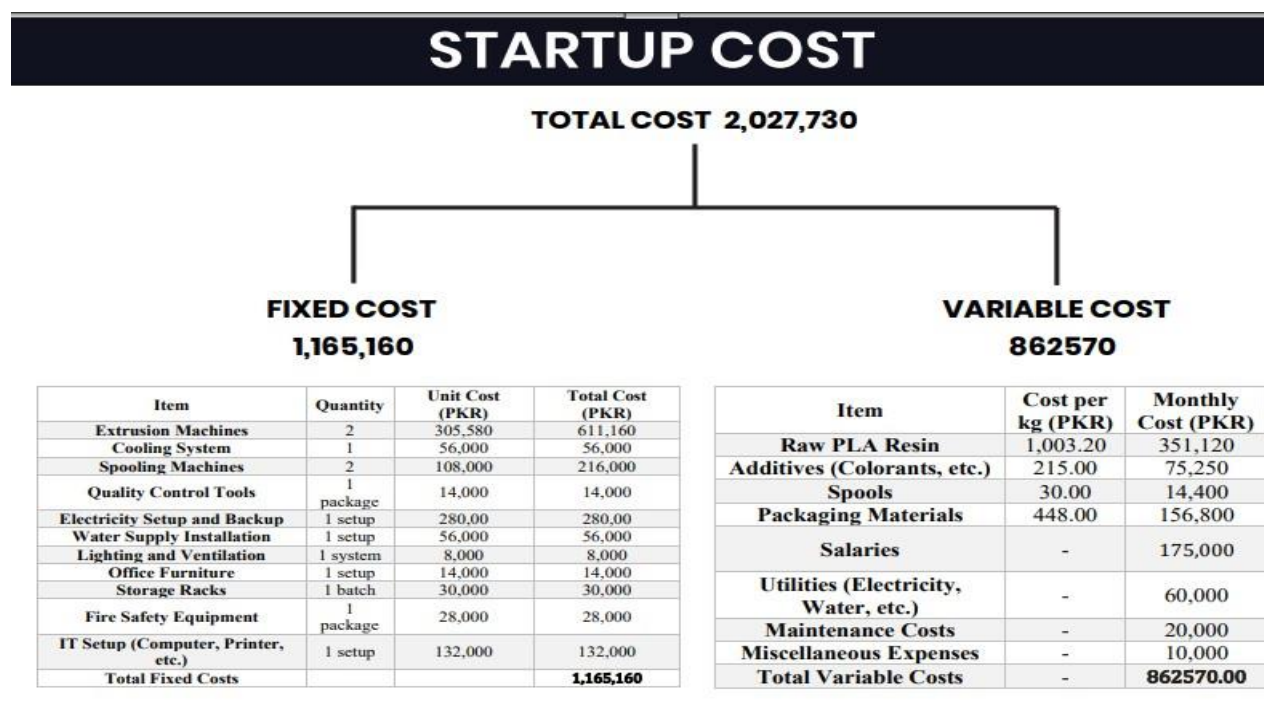


Figure 3: Startup Cost

12.2 Team Résumés and Organizational Chart

Table 3: Organizational Chart

Role	Name	Qualifications	Responsibilities
CEO	Muhammad Huzaifa	B.S. Industrial Engineering	Strategic leadership and operations.
Operations Head	Zain Ul Abideen	B.S. Industrial Engineering	Production planning and supply chain.
Marketing & PR Head	Arslan Ahmad	B.S. Industrial Engineering	Brand building and outreach.
Sales Head	Ali Raza	B.S. Industrial Engineering	Customer acquisition and retention.
Finance Head	Qudrat Ullah	B.S. Industrial Engineering	Budgeting and financial planning.

12.3 Supplier Information

Table 4: Supplier Information

Supplier	Material/Equipment	Location
Suzhou Aifuier Machinery Co.	Extrusion Machines, Cooling System	China
Local Distributors	Raw PLA resin, additives, packaging.	Karachi, Pakistan
Alibaba Suppliers	Specialized PLA materials.	International

12.4 Market Research Data and Growth Projections

Table 5: Growth Projection

Metric	Details
Market Size (Pakistan)	PLA filament demand projected to reach 300 tons/year by 2035.
Growth Rate	CAGR of 19.64% in the global PLA market (Mordor Intelligence).
Target Customers	Universities (NUST, GIKI, UET), small industries, hobbyists.
Competitive Advantage	Locally produced PLA at PKR 5,356/kg compared to imports (PKR 6,000+/kg).
Key Competitors	Imported brands (Creality, eSUN); local suppliers (Print 3D Pakistan, Vertex 3D).

13 Exit Strategy

To ensure long-term sustainability and growth, the proposed exit strategy involves transitioning into a 3D printing company. This pivot leverages the existing setup, aligns with market trends, and taps into high-value product opportunities. Key components of this strategy include:

1. Market Demand:

The growing need for 3D-printed items across industries such as healthcare, education, and manufacturing presents a significant opportunity for expansion.

2. Higher Value Products:

By offering custom 3D-printed items, the company can achieve better profit margins compared to raw filament production.

3. Leverage Existing Setup:

The current equipment, materials, and workforce will be repurposed to efficiently support the new operations, minimizing additional costs and ensuring a smooth transition.

This strategic shift not only addresses evolving market needs but also enhances the company's profitability and adaptability in the competitive 3D printing landscape.